

Uncertainty modelling in AI and engineering

Lecture 2: Random sets

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Outline

1 Random sets

- Definitions
- Random closed sets in $\mathbb{R}^p$

2 Practical models

- Bayesian and consonant random sets
- Random intervals

3 Operations on random sets

- Marginalisation and vacuous extension
- Dempster's rule

Motivation

- In the previous lecture, evidence theory was introduced on a **finite frame** Θ .
- Many practical problems involve variables taking values in an **infinite space**, such as $\mathbb{R}^p$.
- In this lecture, we extend evidence theory to infinite frames using the formal notions of a **random set** in an arbitrary frame Θ and **random closed set** in $\mathbb{R}^p$.

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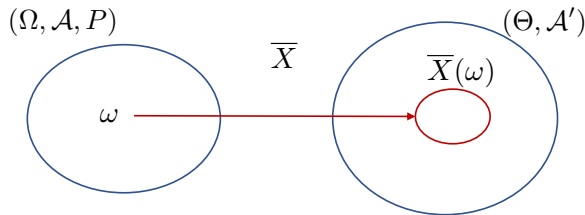
Reminder: probability space

- A **probability space** is a triple $(\Omega, \mathcal{A}, P)$, where
 - ▶ Ω is the **sample space**: the set of possible elementary outcomes;
 - ▶ $\mathcal{A}$ is a **σ -algebra** of subsets of Ω : a collection of subsets of Ω containing Ω , closed under complementation and countable union.
 - ▶ $P : \mathcal{A} \rightarrow [0, 1]$ is a **probability measure**, i.e., a function such that $P(\Omega) = 1$ and, for any sequence $(A_n)_{n \geq 1}$ of pairwise disjoint events in $\mathcal{A}$,

$$P\left(\bigcup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} P(A_n).$$

- A probability space $(\Omega, \mathcal{A}, P)$ is **complete** if $\mathcal{A}$ contains every subset of every null set. (Any probability space can be completed.)
- The pair $(\Omega, \mathcal{A})$ is called a **measurable space**.

Set-valued mapping



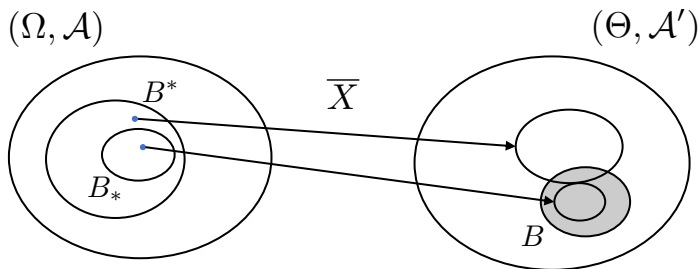
In evidence theory, a set-valued mapping

$$\bar{X} : \Omega \rightarrow \mathcal{P}(\Theta)$$

represents a **piece of evidence** about an unknown X with value in Θ :

- Ω is a set of interpretations of the evidence;
- If interpretation $\omega \in \Omega$ holds, we know that $X \in \bar{X}(\omega)$, and nothing more;
- For any $A \in \mathcal{A}$, $P(A)$ is the probability that the true interpretation lies in A .

Lower and upper inverses



Any $B \subseteq \Theta$ has a **lower inverse** and an **upper inverse**.

Lower inverse

$$B_* = \{\omega \in \Omega : \bar{X}(\omega) \neq \emptyset, \bar{X}(\omega) \subseteq B\}.$$

Set of interpretations that imply $X \in B$.

Upper inverse

$$B^* = \{\omega \in \Omega : \bar{X}(\omega) \cap B \neq \emptyset\}.$$

Set of interpretations compatible with $X \in B$.

Strong measurability

Definition

The set-valued mapping $\bar{X}$ is **strongly measurable** if

$$\forall B \in \mathcal{A}', \quad B^* \in \mathcal{A}.$$

Equivalently, $B_* \in \mathcal{A}$ for every $B \in \mathcal{A}'$.

This is the key technical condition: it ensures that lower and upper inverses of measurable subsets of Θ have probabilities.

Then we can define

$$\text{Bel}_{\bar{X}}(B) = P(B_*), \quad \text{Pl}_{\bar{X}}(B) = P(B^*).$$

Belief, plausibility

For a strongly measurable $\bar{X}$ with $P(\Theta^*) = 1$,

$$\text{Bel}_{\bar{X}}(B) = P\{\bar{X} \subseteq B\}, \quad \text{Pl}_{\bar{X}}(B) = P\{\bar{X} \cap B \neq \emptyset\}.$$

- $\text{Bel}_{\bar{X}}$ is a **belief function**, i.e., $\text{Bel}_{\bar{X}}(\emptyset) = 0$, $\text{Bel}_{\bar{X}}(\Theta) = 1$, and $\text{Bel}_{\bar{X}}$ is **completely monotone**: for any $k \geq 2$ and any collection $B_1, \dots, B_k$ of elements of $\mathcal{A}'$,

$$\text{Pl}\left(\bigcap_{i=1}^k B_i\right) \leq \sum_{\emptyset \neq I \subseteq \{1, \dots, k\}} (-1)^{|I|+1} \text{Pl}\left(\bigcup_{i \in I} B_i\right).$$

- $\text{Pl}_{\bar{X}}$ is the **dual plausibility function**:

$$\text{Pl}_{\bar{X}}(B) = 1 - \text{Bel}_{\bar{X}}(B^c).$$

- Contour function:

$$\text{pl}_{\bar{X}}(\theta) = \text{Pl}_{\bar{X}}(\{\theta\}) = P\{\theta \in \bar{X}\}.$$

Random set: the working definition

Random set

A **random set** is a tuple

$$\mathfrak{R} = (\Omega, \mathcal{A}, P, \Theta, \mathcal{A}', \overline{X}),$$

where $\overline{X} : \Omega \rightarrow \mathcal{P}(\Theta)$ is strongly measurable and $P(\{\omega : \overline{X}(\omega) = \emptyset\}) = 0$. We sometimes call $\overline{X}$ itself a random set.

In practice, we often have $\Theta = \mathbb{R}^p$. We then focus on a particular family of random sets with special properties: **random closed sets**.

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Reminder: closed sets

Open and closed sets

- $G \subseteq \mathbb{R}^p$ is **open** if, around every point of G , some ball of positive radius is entirely contained in G .
- $F \subseteq \mathbb{R}^p$ is **closed** if its complement F^c is open or, equivalently, if the limit of any convergent sequence of points of F still belongs to F .
- $[a, b]$ is closed, $]a, b[$ is open, $[a, b[$ is neither; $\emptyset$ and $\mathbb{R}^p$ are both.
- Closed does not mean bounded: $[a, +\infty[$ is closed. (Closed *and* bounded = **compact**.)
- Singletons, closed balls and boxes, and the level sets $\{\mathbf{x} : g(\mathbf{x}) \geq \omega\}$ of a continuous g are closed.

Reminder: Borel sets in $\mathbb{R}^p$

Borel σ -algebra

$\mathcal{B}(\mathbb{R}^p)$ is the **smallest σ -algebra containing all open subsets** of $\mathbb{R}^p$.

- It contains all closed sets, all intervals, boxes and balls, and everything obtained from them by countable unions, intersections and complements.
- Probability measures on $\mathbb{R}^p$ are defined on $\mathcal{B}(\mathbb{R}^p)$, not on $\mathcal{P}(\mathbb{R}^p)$: this is why we take $\mathcal{A}' = \mathcal{B}(\mathbb{R}^p)$ as our σ -algebra of events.

Random closed set

From now on we take $\Theta = \mathbb{R}^p$, with $\mathcal{A}' = \mathcal{B}(\mathbb{R}^p)$ the Borel σ -algebra.

- Evidence about a physical quantity in $\mathbb{R}^p$ can almost always be described by **closed** focal sets: points, closed intervals, boxes, balls, ellipsoids, level sets, etc.
- Closed-valuedness buys us two decisive advantages: the operations introduced later—projection, cylindrical extension, intersection—**preserve** strong measurability.

Random closed set

A random set $(\Omega, \mathcal{A}, P, \mathbb{R}^p, \mathcal{B}(\mathbb{R}^p), \bar{X})$ such that $\bar{X}(\omega)$ is closed for every $\omega \in \Omega$.

Existence of the commonality function

Define the **commonality function** as

$$Q_{\overline{X}}(B) = P(B^\circ) \quad \text{with} \quad B^\circ := \{\omega \in \Omega : \overline{X}(\omega) \supseteq B\}.$$

This requires $B^\circ \in \mathcal{A}$, which does **not** follow from strong measurability in general.

Theorem

If $\overline{X}$ is a random closed set in $\mathbb{R}^p$ and $(\Omega, \mathcal{A}, P)$ is complete, then $B^\circ \in \mathcal{A}$ for every $B \in \mathcal{B}(\mathbb{R}^p)$: the commonality function $Q_{\overline{X}}$ exists.

All the models of the next section therefore have a commonality function, which sometimes makes Dempster's rule easier to compute.

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Need for practical models

- In most cases, we will use random sets to define **belief functions on $\mathbb{R}^p$** , representing evidence about, e.g., physical parameters such as a distance, a load, a material property, etc.
- We need simple models that can be easily defined, interpreted and combined (see infra).
- We define such models in this section.

Bayesian random sets

Let $X : \Omega \rightarrow \Theta$ be an $(\mathcal{A}, \mathcal{A}')$ measurable function, and set $\overline{X}(\omega) = \{X(\omega)\}$.

- For any $B \in \mathcal{A}'$,

$$B^* = \{\omega : \{X(\omega)\} \cap B \neq \emptyset\} = X^{-1}(B) = B_* \in \mathcal{A},$$

so $\overline{X}$ is strongly measurable, and

$$\text{Bel}_{\overline{X}} = \text{Pl}_{\overline{X}} = P_X,$$

the probability distribution of X . Such a random set is called **Bayesian**.

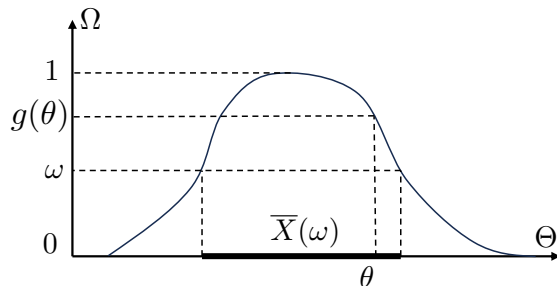
Probability theory is the special case of point-valued evidence

In a metric space, singletons are closed: a Bayesian random set is a random closed set. We identify $\overline{X}$ with X and speak of “the Bayesian random set X with density f ”.

On $\mathbb{R}^p$, a Bayesian random set describes evidence of **infinite precision**: rarely a faithful model of a piece of evidence, but it does arise naturally (e.g., in statistical models).

Consonant random sets

- Let $g : \Theta \rightarrow [0, 1]$ with $\sup_{\theta} g(\theta) = 1$.
- Let $\Omega = [0, 1]$, $\mathcal{A}$ be the Borel σ -algebra $\mathcal{B}([0, 1])$, and λ the Lebesgue (uniform) probability measure on $[0, 1]$.



- The set-valued function

$$\bar{X}(\omega) = \{\theta : g(\theta) \geq \omega\}$$

is strongly measurable.

- The tuple $([0, 1], \mathcal{B}([0, 1]), \lambda, \Theta, \mathcal{P}(\Theta), \bar{X})$ is a random set. Its focal sets are **nested**.

Associated set functions

For the consonant model generated by g ,

$$\text{Pl}_{\bar{X}}(B) = \sup_{\theta \in B} g(\theta), \quad \text{Bel}_{\bar{X}}(B) = 1 - \sup_{\theta \notin B} g(\theta).$$

Moreover,

$$\text{pl}_{\bar{X}}(\theta) = g(\theta), \quad Q_{\bar{X}}(B) = \inf_{\theta \in B} g(\theta).$$

Properties

$\text{Pl}_{\bar{X}}$ is a **possibility measure** and $\text{Bel}_{\bar{X}}$ the dual **necessity measure**:

$$\text{Pl}_{\bar{X}}\left(\bigcup_{i \in I} B_i\right) = \sup_{i \in I} \text{Pl}_{\bar{X}}(B_i), \quad \text{Bel}_{\bar{X}}(A \cap B) = \min\{\text{Bel}_{\bar{X}}(A), \text{Bel}_{\bar{X}}(B)\}.$$

This is the continuous analogue of a consonant mass function.

Consonant random closed sets

- g is **upper semicontinuous** (u.s.c.) if its level sets

$${}^{\omega}g := \{\mathbf{x} \in \mathbb{R}^p : g(\mathbf{x}) \geq \omega\}.$$

are closed for every $\omega \in]0, 1]$.

- The random set $\overline{X}$ is then a **consonant random closed set** in $\mathbb{R}^p$: its focal sets are closed and nested.

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Random closed intervals

Let U and V be random variables such that $U \leq V$ almost surely, and define

$$\bar{X} = [U, V].$$

Theorem

$\bar{X}$ is strongly measurable: $(\Omega, \mathcal{A}, P, \mathbb{R}^p, \mathcal{B}(\mathbb{R}), \bar{X})$ is a random closed set.

Lower and upper cumulative distribution functions (CDFs)

$$F_{\bar{X}}(x) = \text{Bel}_{\bar{X}}([-\infty, x]) = P(V \leq x) = F_V(x),$$

$$\bar{F}_{\bar{X}}(x) = \text{Pl}_{\bar{X}}([-\infty, x]) = P(U \leq x) = F_U(x).$$

Lower and upper expectations

$$\underline{\mathbb{E}}(\bar{X}) = \mathbb{E}(U), \quad \bar{\mathbb{E}}(\bar{X}) = \mathbb{E}(V).$$

Belief, plausibility and commonality of intervals

- For the random interval $\bar{X} = [U, V]$, with $U \leq V$, the belief, plausibility and commonality of a closed interval $[x, y]$ are obtained as follows:

$$\text{Bel}_{\bar{X}}([x, y]) = P([U, V] \subseteq [x, y]) = P(U \geq x, V \leq y),$$

$$\text{Pl}_{\bar{X}}([x, y]) = P([U, V] \cap [x, y] \neq \emptyset) = F_U(y) - F_V(x) + P(V = x),$$

$$Q_{\bar{X}}([x, y]) = P([U, V] \supseteq [x, y]) = P(U \leq x, V \geq y).$$

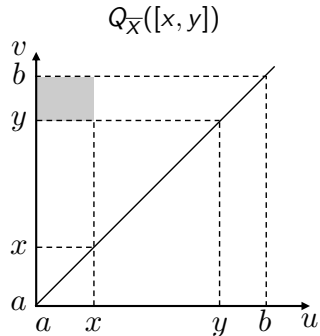
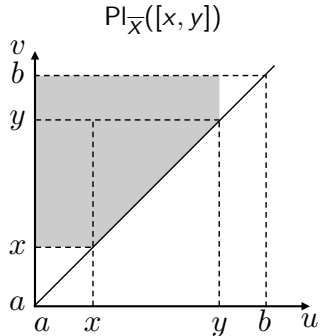
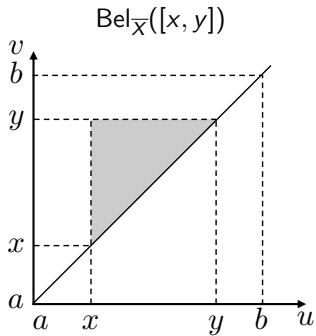
- If (U, V) is absolutely continuous with PDF $f(u, v)$, we have, for instance,

$$Q_{\bar{X}}([x, y]) = \int_{-\infty}^x \int_y^{+\infty} f(u, v) dv du,$$

and, conversely,

$$f(u, v) = - \left. \frac{\partial^2 Q_{\bar{X}}([x, y])}{\partial x \partial y} \right|_{x=u, y=v}.$$

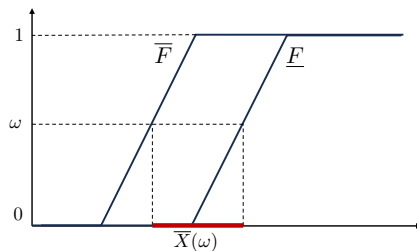
Visualisation



Probability boxes

A **p-box** is a pair of CDFs $(\underline{F}, \overline{F})$ such that

$$\underline{F}(x) \leq \overline{F}(x), \quad x \in \mathbb{R}.$$

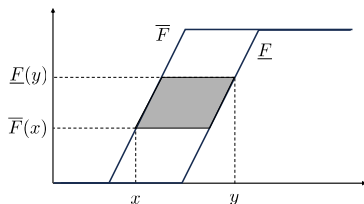


Assuming $\underline{F}$ and $\overline{F}$ continuous, define the source probability space $(]0, 1[, \mathcal{B}(]0, 1[), \lambda)$, where λ is the Lebesgue measure, and the random closed interval

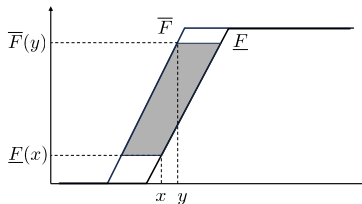
$$\overline{X}(\omega) = [\overline{F}^{-1}(\omega), \underline{F}^{-1}(\omega)], \quad \omega \in]0, 1[.$$

Practical formulas

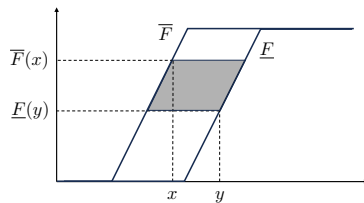
$$\text{Bel}_{\bar{X}}([x, y]) = (\underline{F}(y) - \bar{F}(x))_+$$



$$\text{Pl}_{\bar{X}}([x, y]) = \bar{F}(y) - \underline{F}(x)$$



$$Q_{\bar{X}}([x, y]) = (\bar{F}(x) - \underline{F}(y))_+$$



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Reminder: projection and cylindrical extension

Let $X_1, \dots, X_n$ be unknowns, each X_j with values in $(\Theta_j, \mathcal{A}'_j)$, and let

$$\Theta = \Theta_1 \times \dots \times \Theta_n, \quad \Theta_I = \prod_{i \in I} \Theta_i, \quad \mathcal{A}'_I = \bigotimes_{i \in I} \mathcal{A}'_i,$$

for $\emptyset \neq I \subseteq \{1, \dots, n\}$, and write $J = I^c$.

Cylindrical extension and projection

$$\text{cyl}_I^\Theta(A_I) = A_I \times \Theta_J \quad (A_I \subseteq \Theta_I), \quad \text{proj}_I(B) = \{x_I \in \Theta_I : \exists x \in B \text{ with } I\text{-components } x_I\}.$$

The mapping cyl_I^Θ is a **refining** of Θ_I in Θ , and $\text{cyl}_I^\Theta = \text{proj}_I^{-1}$. Its **upper inverse** is proj_I .

Marginalisation

Marginal

The **marginal** in Θ_I of a random set $\mathfrak{R} = (\Omega, \mathcal{A}, P, \Theta, \mathcal{A}', \bar{X}^\Theta)$ is

$$\mathfrak{R}^{\downarrow \Theta_I} = \left(\Omega, \mathcal{A}, P, \Theta_I, \mathcal{A}'_I, \bar{X}^{\Theta \downarrow \Theta_I} \right), \quad \bar{X}^{\Theta \downarrow \Theta_I} := \text{proj}_I \circ \bar{X}^\Theta.$$

It is **always** a random set: no additional condition is required.

Proposition

For all $B \in \mathcal{A}'_I$,

$$\text{Pl}_{\bar{X}^{\Theta \downarrow \Theta_I}}(B) = \text{Pl}_{\bar{X}^\Theta} \left(\text{cyl}_I^\Theta(B) \right), \quad \text{Bel}_{\bar{X}^{\Theta \downarrow \Theta_I}}(B) = \text{Bel}_{\bar{X}^\Theta} \left(\text{cyl}_I^\Theta(B) \right).$$

Example: the marginal in Θ_i of a random ball with center $\mathbf{M} \sim N(\boldsymbol{\mu}, \mathbf{I})$ and constant radius r is the random interval $[M_i - r, M_i + r]$, with $M_i \sim N(\mu_i, 1)$.

Vacuous extension

Conversely, let $\mathfrak{R}^{\Theta_I} = (\Omega, \mathcal{A}, P, \Theta_I, \mathcal{A}', \bar{X})$ and put

$$\bar{X}^{\uparrow\Theta} := \text{cyl}_I^{\Theta} \circ \bar{X}, \quad \bar{X}^{\uparrow\Theta}(\omega) = \bar{X}(\omega) \times \Theta_J.$$

Warning: unlike marginalisation, this requires an assumption— $\bar{X}^{\uparrow\Theta}$ need not be strongly measurable in general.

Theorem

If $\Theta = \mathbb{R}^p$, $\bar{X}$ is closed-valued and $(\Omega, \mathcal{A}, P)$ is complete, then $\bar{X}^{\uparrow\Theta}$ is strongly measurable, the **vacuous extension** $\mathfrak{R}^{\uparrow\Theta}$ is well defined, and, for all $B \in \mathcal{A}'$,

$$\text{Pl}_{\bar{X}^{\uparrow\Theta}}(B) = \text{Pl}_{\bar{X}}(\text{proj}_I(B)), \quad \text{Bel}_{\bar{X}^{\uparrow\Theta}}(B) = \text{Bel}_{\bar{X}}\left(\{x_I : \text{cyl}_I^{\Theta}(\{x_I\}) \subseteq B\}\right).$$

Example: the vacuous extension of the random interval $[U, V]$ in $\mathbb{R}^2$ is the random **strip**

$$\bar{X}^{\uparrow\mathbb{R}^2}(\omega) = [U(\omega), V(\omega)] \times \mathbb{R}.$$

Marginalisation loses information

For every ω ,

$$\overline{X}(\omega) \subseteq \text{cyl}_I^\Theta (\text{proj}_I (\overline{X}(\omega))) = \left(\overline{X}^{\downarrow \Theta_I} \right)^{\uparrow \Theta} (\omega),$$

with equality if and only if $\overline{X}(\omega)$ is a cylinder.

Consequence

For all $B \in \mathcal{A}'$,

$$\text{Bel}_{(\overline{X}^{\downarrow \Theta_I})^{\uparrow \Theta}}(B) \leq \text{Bel}_{\overline{X}}(B) \quad \text{and} \quad \text{Pl}_{(\overline{X}^{\downarrow \Theta_I})^{\uparrow \Theta}}(B) \geq \text{Pl}_{\overline{X}}(B).$$

Projecting and re-extending yields a **less committed** random set: an outer approximation.

- Marginalisation is **lossy**, exactly as in probability theory.
- Vacuous extension adds **no** information: it only re-expresses evidence about X_I as evidence about X .

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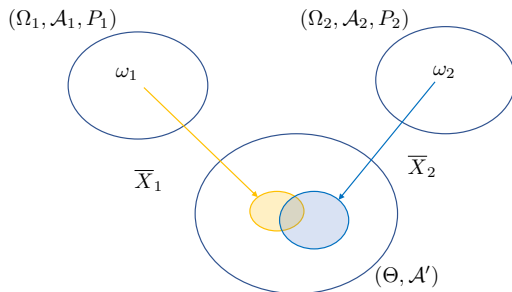
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Dempster's rule: the idea

Two **independent** pieces of evidence about the same unknown X :

$$\mathfrak{R}_i = (\Omega_i, \mathcal{A}_i, P_i, \Theta, \mathcal{A}', \bar{X}_i), \quad i = 1, 2.$$



- Interpretations are drawn independently:
 $(\omega_1, \omega_2) \sim P_1 \otimes P_2$.
- If ω_1 and ω_2 both hold, then

$$X \in \bar{X}_1(\omega_1) \cap \bar{X}_2(\omega_2).$$

- An **empty** intersection means that the two interpretations **contradict** each other: such a pair must be ruled out.

Dempster's rule = **intersect**, then **condition on non-contradiction**.

Conflict and combinability

Define $\overline{X}_1 \cap \overline{X}_2 : (\omega_1, \omega_2) \mapsto \overline{X}_1(\omega_1) \cap \overline{X}_2(\omega_2)$ and let

$$\Theta^* = \{(\omega_1, \omega_2) \in \Omega_1 \times \Omega_2 : \overline{X}_1(\omega_1) \cap \overline{X}_2(\omega_2) \neq \emptyset\}$$

be the set of **non-contradictory** pairs of interpretations. The **degree of conflict** is

$$\kappa = 1 - (P_1 \otimes P_2)(\Theta^*).$$

$\overline{X}_1$ and $\overline{X}_2$ are **combinable** if

C_1 : $\kappa < 1$: the two pieces of evidence are not totally contradictory;

C_2 : $\overline{X}_1 \cap \overline{X}_2$ is strongly measurable w.r.t. $\mathcal{A}_1 \otimes \mathcal{A}_2$ and $\mathcal{A}'$.

When does C_2 hold?

Always, for random closed sets in $\mathbb{R}^p$ with a complete product probability space. Only C_1 has to be checked in practice.

The orthogonal sum

If $\overline{X}_1$ and $\overline{X}_2$ are combinable, their **orthogonal sum** is the random set

$$\mathfrak{R}_1 \oplus \mathfrak{R}_2 := (\Omega_1 \times \Omega_2, \mathcal{A}_1 \otimes \mathcal{A}_2, P_{12}^*, \Theta, \mathcal{A}', \overline{X}_1 \cap \overline{X}_2),$$

where $P_{12}^* := (P_1 \otimes P_2)(\cdot \mid \Theta^*)$ is the product measure **conditioned on non-contradiction**. We write $\overline{X}_1 \oplus \overline{X}_2$ for short.

Induced set functions (with $P := P_1 \otimes P_2$)

$$\text{Bel}_{\overline{X}_1 \oplus \overline{X}_2}(B) = P(\overline{X}_1 \cap \overline{X}_2 \subseteq B \mid \overline{X}_1 \cap \overline{X}_2 \neq \emptyset),$$

$$\text{Pl}_{\overline{X}_1 \oplus \overline{X}_2}(B) = P(\overline{X}_1 \cap \overline{X}_2 \cap B \neq \emptyset \mid \overline{X}_1 \cap \overline{X}_2 \neq \emptyset),$$

$$Q_{\overline{X}_1 \oplus \overline{X}_2}(B) = P(\overline{X}_1 \cap \overline{X}_2 \supseteq B \mid \overline{X}_1 \cap \overline{X}_2 \neq \emptyset).$$

Properties of Dempster's rule

Algebraic properties

Neutral element: the **vacuous** random set $\bar{X}_0 \equiv \Theta$ satisfies $\bar{X} \oplus \bar{X}_0 = \bar{X}_0 \oplus \bar{X} = \bar{X}$;

Commutativity: $\bar{X}_1 \oplus \bar{X}_2 = \bar{X}_2 \oplus \bar{X}_1$;

Associativity: $(\bar{X}_1 \oplus \bar{X}_2) \oplus \bar{X}_3 = \bar{X}_1 \oplus (\bar{X}_2 \oplus \bar{X}_3)$,

up to the canonical identification of the product sample spaces.

- Consequence: n pieces of evidence may be combined **in any order**, and we may write $\bigoplus_{k=1}^n \bar{X}_k$ unambiguously.
- The **weight of conflict** $\text{Con} = -\ln(1 - \kappa)$ is **additive**:

$$\text{Con}(\bar{X}_1, \dots, \bar{X}_n) = \text{Con}(\bar{X}_1, \dots, \bar{X}_{n-1}) + \text{Con}(\bar{X}_1 \oplus \dots \oplus \bar{X}_{n-1}, \bar{X}_n).$$

Commonality and contour functions

Theorem

If $\overline{X}_1$ and $\overline{X}_2$ are combinable and have commonality functions Q_1 and Q_2 , then, for every nonempty $B \in \mathcal{A}'$,

$$Q_{\overline{X}_1 \oplus \overline{X}_2}(B) = \frac{Q_1(B) Q_2(B)}{1 - \kappa}.$$

Proof: for $B \neq \emptyset$, $\{\overline{X}_1 \cap \overline{X}_2 \supseteq B\} = \{\overline{X}_1 \supseteq B\} \times \{\overline{X}_2 \supseteq B\}$, and $P_1 \otimes P_2$ factorises on product events.

Corollary: contour functions multiply

$$pl_{\overline{X}_1 \oplus \overline{X}_2}(\theta) = \frac{pl_1(\theta) pl_2(\theta)}{1 - \kappa}.$$

- Combination reduces to a **pointwise product**: the cheapest possible computation.
- Since a contour function is bounded by 1, $1 - \kappa \geq \sup_{\theta} pl_1(\theta) pl_2(\theta)$.

Conditioning as a special case

Let $B \in \mathcal{A}'$, $B \neq \emptyset$, and let $\bar{X}_B$ be the **constant** random set $\bar{X}_B(\omega_2) = B$ for all ω_2 .

- $\bar{X} \cap \bar{X}_B$ is always strongly measurable, since $(\bar{X} \cap \bar{X}_B)^*(C) = \bar{X}^*(B \cap C) \times \Omega_2$;
- $(P_1 \otimes P_2)(\Theta^*) = \text{Pl}_{\bar{X}}(B)$, so $\bar{X}$ and $\bar{X}_B$ are combinable iff $\text{Pl}_{\bar{X}}(B) > 0$.

Dempster's conditioning

$$\text{Pl}_{\bar{X}}(A \mid B) = \frac{\text{Pl}_{\bar{X}}(A \cap B)}{\text{Pl}_{\bar{X}}(B)}, \quad \text{Bel}_{\bar{X}}(A \mid B) = \frac{\text{Bel}_{\bar{X}}(A \cup B^c) - \text{Bel}_{\bar{X}}(B^c)}{1 - \text{Bel}_{\bar{X}}(B^c)}.$$

- Exactly the formulas obtained in the finite case.
- If $\bar{X}$ is Bayesian, both reduce to **Bayes' conditioning** $P(A \mid B) = P(A \cap B)/P(B)$.

Example: two one-sided pieces of evidence

Let $X_1 \sim N(\mu_1, \sigma_1^2)$ and $X_2 \sim N(\mu_2, \sigma_2^2)$ be independent, and consider

$$\overline{X}_1 = [X_1, +\infty[\quad ("X \geq X_1"), \quad \overline{X}_2 =]-\infty, X_2] \quad ("X \leq X_2").$$

For $x \leq y$, $Q_1([x, y]) = \Phi\left(\frac{x-\mu_1}{\sigma_1}\right)$ and $Q_2([x, y]) = 1 - \Phi\left(\frac{y-\mu_2}{\sigma_2}\right)$, while

$$\kappa = P(X_1 > X_2) = \Phi\left(\frac{\mu_1 - \mu_2}{\sqrt{\sigma_1^2 + \sigma_2^2}}\right).$$

Multiplying the commonalities and differentiating twice, $\overline{X}_1 \oplus \overline{X}_2$ is the random interval $[U, V]$ with joint density

$$f(u, v) = \frac{\sigma_1^{-1} \sigma_2^{-1} \phi\left(\frac{u-\mu_1}{\sigma_1}\right) \phi\left(\frac{v-\mu_2}{\sigma_2}\right)}{\Phi\left(\frac{\mu_2 - \mu_1}{\sqrt{\sigma_1^2 + \sigma_2^2}}\right)} \mathbb{1}_{[0, +\infty[}(v - u),$$

i.e., the distribution of (X_1, X_2) **conditioned on $X_1 \leq X_2$** .

Combination in product frames

Pieces of evidence often bear on **different** variables. Let $\bar{X}^{\Theta_I}$ and $\bar{X}^{\Theta_J}$ be random closed sets, $I, J \subseteq \{1, \dots, n\}$: we combine them in the smallest common frame $\Theta_{I \cup J}$,

$$\bar{X}^{\Theta_I} \oplus \bar{X}^{\Theta_J} := \bar{X}^{\Theta_I \uparrow \Theta_{I \cup J}} \oplus \bar{X}^{\Theta_J \uparrow \Theta_{I \cup J}}.$$

- Both vacuous extensions exist and their intersection is strongly measurable: only $\kappa < 1$ has to be checked.
- If $I \cap J = \emptyset$, then $\kappa = 0$: evidence about **different** variables never conflicts.

Random boxes

Combining p independent random intervals $[U_i, V_i]$ yields the **random box**

$$\bar{X}(\omega_1, \dots, \omega_p) = \prod_{i=1}^p [U_i(\omega_i), V_i(\omega_i)].$$

Boxes are stable under intersection, hence under Dempster's rule—random balls are not.

Monte Carlo combination

To combine n random sets when closed forms are unavailable:

- 1 Draw $(\omega_1, \dots, \omega_n)$ from the joint law.
- 2 Compute $F = \bigcap_{k=1}^n \overline{X}_k(\omega_k)$.
- 3 If $F = \emptyset$, reject the draw.
- 4 Otherwise, store F as a focal set of the combined random set.
- 5 Repeat until N focal sets have been accepted.

$$\hat{\kappa} = \frac{\# \text{ rejected draws}}{\# \text{ total draws}}.$$

The procedure becomes inefficient when κ is close to 1; importance sampling or MCMC can then be used.

Estimating belief and plausibility

After simulation, approximate the combined random set by

$$m(F_i) = \frac{1}{N}, \quad i = 1, \dots, N.$$

For any event $B \subseteq \Theta$,

$$\widehat{\text{Bel}}(B) = \frac{1}{N} \#\{i : F_i \subseteq B\},$$

$$\widehat{\text{Pl}}(B) = \frac{1}{N} \#\{i : F_i \cap B \neq \emptyset\}.$$

Estimated standard errors:

$$\text{se}(\widehat{p}) = \sqrt{\frac{\widehat{p}(1 - \widehat{p})}{N}}.$$

Summary

- A **random set** is a strongly measurable set-valued mapping; it induces $\text{Bel}_{\bar{X}}(B) = P(\bar{X} \subseteq B)$ and $\text{Pl}_{\bar{X}}(B) = P(\bar{X} \cap B \neq \emptyset)$ on an infinite frame.
- In $\mathbb{R}^p$, **closed** focal sets on **complete** probability spaces make everything work: commonality functions exist, and intersections and cylindrical extensions stay measurable.
- Practical models: **Bayesian** (= probability), **consonant** random sets (= possibility), **random intervals** and **p-boxes**, **random balls** and **boxes**.
- **Marginalisation** is always defined and lossy; **vacuous extension** needs closedness and completeness, and adds no information.
- **Dempster's rule** = intersect and condition on non-contradiction. Commonality and contour functions simply **multiply**.
- Next: **random fuzzy sets**, replacing focal sets by focal fuzzy sets (i.e., u.s.c. functions).

Further reading

cf. <http://www.hds.utc.fr/~tdenoeux>



I. Molchanov

Theory of Random Sets.

Springer, 2005.



T. Denœux

Modern Evidence Theory: A Unified Approach to Uncertainty in AI, Engineering and Statistics.

Book manuscript (forthcoming).